

Evaluation Choices Shape Apparent Quantum Kernel Robustness under Distribution Shift

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Abstract

Claims of quantum-kernel robustness can depend on how candidates are tuned, selected, and compared. We audit this dependence with a controlled kernel-swap benchmark under distribution shift and prospectively frozen external corroboration. In eight fixed security scenario-groups, same-test target-label selection, fixed regularization, and customary references produce apparent gains up to $+0.037$. Within one generation, moving from this bundle to train-only tuning, ID-validation selection, extended references, and matched counts changes the source-dataset-equal support-vector effect from $+0.0131$ to -0.0047 . Under full controls it is -0.0056 , and -0.0009 for Gaussian-process classification. Two of four quantum maps are separable product kernels. At matched 30-candidate budgets, entangling-ZZ and product-map support-vector effects are -0.0093 and -0.0055 . Across three fixed external tasks, the controlled support-vector effect is -0.0119 . Neither stronger controls nor restriction to entangling maps reveals consistent evidence of a positive quantum-family effect in these low-qubit fixed cases.

Keywords: quantum machine learning; quantum kernels; distribution shift; out-of-distribution robustness; kernel methods; kernel-target alignment; Gaussian processes; malware detection; network intrusion detection

047 1 Introduction

048
049 Quantum kernel methods embed classical data into quantum states and use state over-
050 laps as similarities for otherwise classical learners [1–3]. This separation makes them
051 a clean setting for asking whether quantum-induced feature geometry provides useful
052 inductive bias. It also makes comparison unusually sensitive to evaluation design: the
053 downstream learner can be held fixed, but conclusions still depend on the encoding,
054 bandwidth, regularization, classical reference set, and rule used to select a candidate
055 [4–7].

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057 Recent benchmarks have accordingly shifted the question from whether an isolated
058 quantum model can perform well to whether a claimed gain survives strong and sym-
059 metric controls. Broad QML studies find competitive classical baselines once pipelines
060 are aligned [6]; quantum-kernel benchmarks show substantial sensitivity to feature
061 map and hyperparameter choice [7]; and a concurrent tabular study using nested cross-
062 validation, spectral analysis, and hardware validation finds no significant quantum
063 win across its tested pairs [8]. Out-of-distribution generalization in QML is itself an
064 established topic: prior work analyzes learning quantum dynamics outside the training
065 distribution and relates data-dependent generalization to quantum Fisher information
066 [9, 10]. Our narrower deployment question is whether a family-level quantum-kernel
067 robustness claim on classical data survives target-label-free selection and symmetric
068 tuning, reference, and search-budget controls.

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070 Distribution shift makes model selection part of the estimand. Work on domain
071 generalization and real-world shift has shown that robustness gains can disappear
072 when search budgets, hyperparameters, and selection criteria are standardized [11–14].
073 A selector that inspects the shifted test labels estimates an oracle upper bound, not
074 deployable performance; a family with more candidates has more opportunities to win
075 that selection. A controlled quantum–classical evaluation under shift must therefore

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hold fixed not only the classifier and data split, but also regularization, candidate budget, and label access.

Security detection provides a useful stress test because shift is intrinsic to the application. Malware evolves over time, and experimental bias can overstate detector performance [15–17]; network traffic changes with infrastructure and attack campaigns [18, 19]. Quantum classifiers have already been evaluated on intrusion-detection benchmarks, sometimes with reported gains over small classical reference sets on i.i.d. splits [20–22]. We use these datasets not to propose a security system, but to expose kernel families to reproducible constructed shifts and to a held-out attack-campaign shift.

The closest kernel-robustness-specific concurrent work studies additive input corruption and proposes a spectral phase encoding [23]. Our question is complementary: rather than optimize a new encoding or claim the first QML study under shift, we audit how evaluation choices change a family-level conclusion under constructed and held-out-campaign shifts. The distinguishing contribution is a controlled classical-data kernel swap combining no-OOD-label selection, equal candidate budgets, per-configuration train-only regularization, a within-generation factorial sensitivity, and prospectively frozen external corroboration.

We implement a controlled *kernel-swap* benchmark in which preprocessing, splits, classifier, and family-internal selection logic are fixed while only the kernel changes. Four security scenarios from EMBER, UNSW-NB15, and ToN-IoT yield eight scenario-groups across constructed and held-out-campaign shifts. Support-vector and Laplace Gaussian-process classifiers consume the same precomputed Gram matrices. The quantum family contains four fidelity feature-map families with angle-scale sweeps; the classical family extends linear and RBF references with polynomial, Laplacian, and Matérn kernels and symmetric bandwidth freedom. Each configuration’s regularization is tuned from training data alone. The primary deployed candidate is selected on a disjoint ID-validation split, and the two extended families receive equal

139 candidate budgets. Fifteen pipeline realizations per setting quantify conditional vari-
140 ability; the benchmarks are treated as fixed case studies rather than as random draws
141 from an undefined population.
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144 The comparison is coupled to a circuit and geometry audit. Two of the four
145 implemented quantum maps contain two-qubit interactions; the other two prepare
146 product states whose fidelity kernels factorize over coordinates. We therefore sepa-
147 rate entangling and product-map results, report logical circuit and shot resources,
148 and restrict the current geometry argument to effective rank and centered kernel-
149 target alignment [24–26]. Complementary work links data-induced randomness to
150 classification through a class-margin diagnostic [27]; here geometry is used to test asso-
151 ciations under shift and family controls, not to introduce a new embedding metric.
152 Partial association, leave-one-source-dataset-out prediction, and effective-rank match-
153 ing constrain interpretation; repeated finite-shot perturbations compare unprojected,
154 independent-square PSD, and coherent train-eigenspace PSD estimators.
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157 The evaluation choices change the conclusion. With same-test oracle selection,
158 fixed regularization, and customary linear+RBF baselines, fidelity kernels appear to
159 improve OOD balanced accuracy by as much as +0.037. With train-only regular-
160 ization, no-OOD-label selection, and structurally matched candidate budgets, the
161 source-dataset-equal quantum-minus-classical delta is -0.0056 for SVC and -0.0009
162 for the GP classifier. Across the fixed cases, we observe no consistent evidence of a
163 quantum advantage under this controlled endpoint. Effective rank and OOD align-
164 ment remain associated with accuracy, but the association is regime-dependent and
165 shared by both families. Nearest-rank pairing within the predefined overlap yields neg-
166 ative median differences in all eight groups, including under one-to-one and caliper
167 sensitivities. Repeated finite-shot estimates inflate exact-statevector effective rank and
168 produce condition-dependent OOD deviations.
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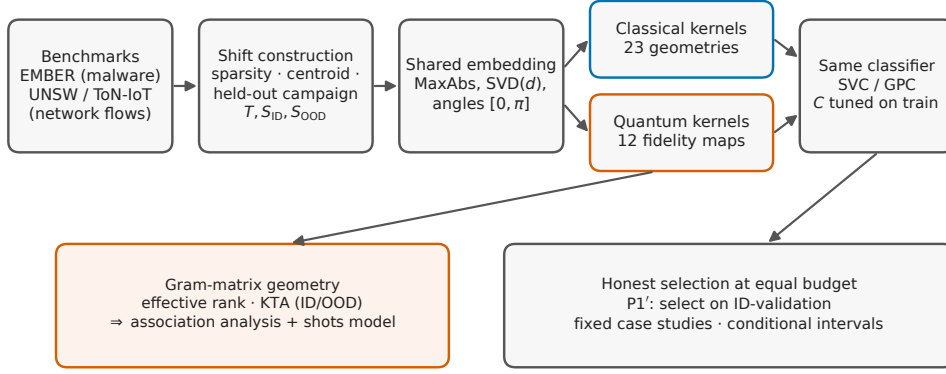


Fig. 1 The controlled kernel-swap protocol. Benchmark pools feed reproducible shift constructions; all kernels operate on a shared train-only embedding, receive per-configuration regularization tuned on the training split, and are consumed by the same precomputed-kernel classifiers. Family-level conclusions are drawn by target-label-free, equal-budget selection on an in-distribution validation split (P1'). Each scenario-group interval is a 95% conditional cluster- t interval over five q-split seed clusters ($n = 5$), conditional on the fixed benchmark pools and design. Gram matrices feed the associational geometry audit and repeated finite-shot sensitivity.

The contribution is therefore the combined evaluation design: distribution shift, target-label-free selection, train-only regularization, structurally matched search budgets, a within-v4 factorial, circuit-aware quantum strata, and prospectively frozen external corroboration. Together these controls show how the reported quantum-kernel robustness result changes across evaluation bundles and whether the observed behaviour requires entangling feature geometry. The scope is deliberately limited to the low-qubit fidelity kernels, datasets, and shift constructions studied here.

2 Results

2.1 Overview

All comparisons share classifier, preprocessing, and split definitions within a scenario-group; only the kernel changes. Each setting contains fifteen pipeline realizations nested in five q-split clusters. We first contrast an optimistic same-test oracle against the controlled endpoint, then separate product and entangling maps and decompose

evaluation axes within v4. Geometry and finite-shot analyses are explicitly diagnostic. Throughout, the eight security groups are fixed cases; pointwise conditional intervals describe pipeline variation given the benchmark pools and design, not uncertainty over a population of datasets.

2.2 The apparent advantage under oracle selection

We begin with a deliberately optimistic reference comparison: fidelity-based quantum kernels against linear and RBF baselines, with each family’s representative chosen by Best-by-OOD selection—the configuration with the highest balanced accuracy on the OOD test split itself—and the classical kernels left at the default regularization $C = 1$. Related intrusion-detection studies motivate the small-reference context [20–22], but we do not claim that this exact protocol dominates QML practice. Under these conventions, on EMBER, the quantum family shows an OOD advantage of +0.037 under SVC and +0.028 under the GP classifier against linear+RBF, shrinking to +0.004/+0.002 once the classical family is enlarged with heavier-tailed kernels. Supplementary Table 5 gives the complete oracle results.

Two features favour this apparent quantum result. Best-by-OOD is an *oracle*: it uses the labels it then reports. Fixing $C = 1$ also leaves classical candidates at a potentially unsuitable regularization point. These values are therefore descriptive upper bounds, not deployable effects.

2.3 A cross-generation sensitivity map is associational

To avoid privileging a preferred endpoint, we enumerated all ten logically valid, already-computed specifications fixed before this extension (Figure 2). They trace a descriptive path over target-label access, SVC regularization, classical reference strength, and candidate budget. S1–S4 use the legacy generation with fixed $C = 1$, whereas S5–S10 use the corrected v4 generation with train-cross-validated C ; the

S4-S5 transition therefore does not isolate regularization from generation. Every specification retains all eight security scenario-groups.

The family-level conclusion spans both sides of zero. For SVC, the source-dataset-equal quantum-minus-classical mean ranges from +0.047 under legacy ID selection with fixed $C = 1$ and the customary reference (S2) to -0.006 under ID-validation selection, train-only regularization, the extended reference, and equal budgets (S10). For the GP classifier, for which the regularization axis is not applicable, the corresponding range is +0.042 to -0.003 . Expanding the classical reference contracts the positive effect in the relevant legacy comparisons; the complete controlled v4 bundle yields estimates at or slightly below zero. The path is not monotone in every intermediate specification, which is precisely why reporting only an oracle and a final controlled endpoint would be insufficient. The curve shows that the apparent advantage is conditional on a bundle of evaluation decisions rather than a stable property of quantum provenance.

2.4 The controlled comparison does not preserve the apparent advantage

A deployable comparison must tune each configuration’s regularization without looking at the test data, select the deployed configuration without OOD labels, and compare families at the same candidate budget. We implement all three (Section 4). For every kernel candidate, the SVC regularization C is chosen by five-fold cross-validation on the training Gram matrix alone. The deployed configuration is then the one with the best balanced accuracy on an in-distribution *validation* split, held out from the ID test split (protocol P1’); no OOD label is ever consulted in selection. The primary comparison uses 60 candidates per extended family in every scenario-group and classifier, with the full 115-candidate classical pool retained as a sensitivity. We treat the eight scenario-groups as fixed case studies and report, per group, the

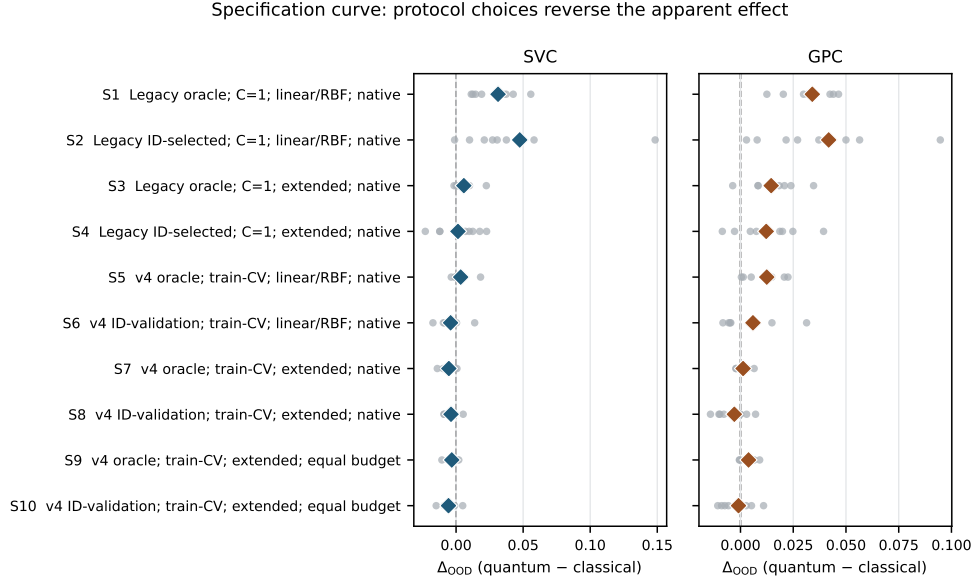


Fig. 2 A prespecified cross-generation sensitivity map shows how the apparent family effect changes across evaluation bundles. Grey points are the eight fixed security scenario-groups; diamonds average groups within EMBER, UNSW-NB15, and ToN-IoT before giving the three source datasets equal weight. Specifications are shown in their prespecified S1–S10 order, not sorted by effect. S1–S4 use the legacy generation with fixed $C = 1$; S5–S10 use the corrected v4 generation with train-cross-validated C , so their boundary is generation-confounded and does not isolate a regularization effect. Positive values favour the fidelity-kernel family. GPC has no C -regularization axis. These are descriptive fixed-case summaries, not per-axis causal effects; no specification-level hypothesis test or population interval is reported.

quantum-minus-classical OOD delta with a conditional interval over the five split realizations, and no population p -value (Section 4).

Table 1 and Figure 3 show no consistent evidence of a quantum advantage under the controlled endpoint. Against the equal-budget extended classical family, the source-dataset-equal difference is -0.00565 for SVC (leave-one-source-dataset-out range $[-0.00702, -0.00450]$) and -0.00094 for GPC ($[-0.00190, +0.00094]$). Against the customary linear+RBF reference, the corresponding differences are -0.00407 and $+0.00591$, with heterogeneous group signs. UNSW-NB15 reconnaissance under held-out-campaign shift has a small positive mean ($+0.005$ for both classifiers), with

Table 1 Target-label-free selection results for eight fixed scenario-groups. Values are quantum-minus-classical OOD balanced-accuracy differences under $P1'$ (select on ID-validation, tune per-configuration C on train, evaluate on OOD test). The frozen extended-family analysis uses 60 candidates per family in every group-classifier cell. Each scenario interval is a 95% conditional cluster- t interval over five q-split seed clusters ($n = 5$), conditional on the benchmark pools and design; the fifteen pipeline realizations per setting are nested within these clusters. An interval containing zero is not an equivalence result. Source-dataset-equal rows average groups within three source datasets and report leave-one-source-dataset-out ranges, not confidence intervals. The customary linear+RBF comparison is contextual rather than equal-budget.

Scenario	Clf.	vs lin./RBF	vs ext. (equal B)	interval / LODO range
EMBER $m1$	SVC	-0.0095	-0.0058	$[-0.0340, +0.0223]$
	GPC	-0.0050	-0.0009	$[-0.0101, +0.0083]$
EMBER $m2$	SVC	+0.0139	-0.0064	$[-0.0137, +0.0009]$
	GPC	+0.0149	+0.0027	$[-0.0027, +0.0081]$
UNSW-DoS (campaign)	SVC	-0.0043	-0.0033	$[-0.0139, +0.0073]$
	GPC	+0.0041	-0.0057	$[-0.0120, +0.0005]$
UNSW-DoS (constructed)	SVC	-0.0085	-0.0044	$[-0.0119, +0.0031]$
	GPC	-0.0056	-0.0108	$[-0.0177, -0.0039]$
UNSW-Recon (campaign)	SVC	+0.0004	+0.0050	$[-0.0058, +0.0157]$
	GPC	+0.0074	+0.0052	$[-0.0019, +0.0123]$
UNSW-Recon (constructed)	SVC	-0.0076	-0.0089	$[-0.0146, -0.0031]$
	GPC	-0.0083	-0.0075	$[-0.0208, +0.0059]$
ToN-IoT (campaign)	SVC	-0.0172	-0.0148	$[-0.0268, -0.0029]$
	GPC	-0.0046	-0.0090	$[-0.0190, +0.0010]$
ToN-IoT (constructed)	SVC	-0.0016	-0.0011	$[-0.0049, +0.0028]$
	GPC	+0.0313	+0.0109	$[+0.0017, +0.0202]$
Source-dataset-equal mean, SVC		-0.0041	-0.0056	$[-0.0070, -0.0045]$
Source-dataset-equal mean, GPC		+0.0059	-0.0009	$[-0.0019, +0.0009]$

pointwise conditional intervals spanning zero. The full-pool extended-family sensitivity is similar in direction and scale (-0.00378 for SVC and -0.00293 for GPC).

The candidate-budget result is insensitive to the sampling scheme.

The primary `kernel.blocked` scheme samples 12 whole classical kernel-shape blocks crossed with all five dimensions, matching both the quantum count and its feature-map-by-dimension structure. Uniform and kernel-stratified samples match the count but not this block structure. The source-dataset-equal resampling estimates remain close under all three schemes: -0.00576 , -0.00555 , and -0.00548 for SVC, and -0.00185 , -0.00232 , and -0.00253 for GPC, respectively. The largest scheme range in any group-classifier cell is 0.00353. Supplementary Table 2 reports all 48 cells.

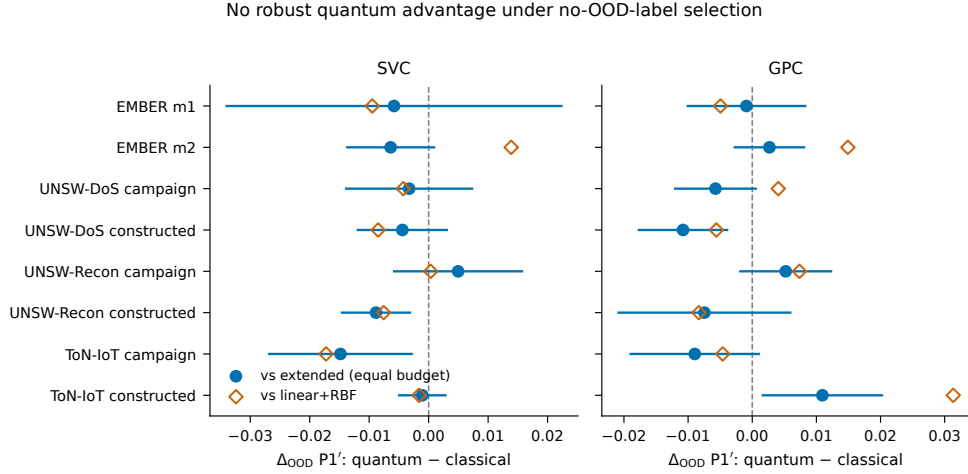


Fig. 3 Controlled target-label-free family effects in eight fixed scenario-groups. P1' OOD differences (quantum minus classical) are shown for SVC (left) and GPC (right). Filled circles compare against the equal-budget extended classical family; error bars are pointwise 95% conditional cluster- t intervals over five q-split clusters ($n = 5$), conditional on the fixed benchmark pools and design. Open diamonds compare against the customary linear+RBF reference. The intervals are not simultaneous and do not imply population inference or equivalence.

Candidate-count equality is therefore distinct from search-space isomorphism, but this distinction does not drive the conclusion here.

Rank-matched differences remain negative within the predefined overlap.

We pair each quantum configuration with the classical configuration of nearest training effective rank within the same run and dimension. This observational matching reuses classical candidates and applies the prespecified caliper $\max(r_Q, r_C) / \min(r_Q, r_C) \leq 1.25$. It retains 48,708 of 64,800 pairs (75.2%; 69.3–83.3% by group). Figure 4 reports negative median quantum-minus-classical differences in all eight groups (−0.0013 to −0.0094); the quantum member is more accurate in 22–37% of retained pairs. The pooled median is −0.0056. Minimum-cost one-to-one matching retains 24,675 pairs at the same caliper and yields a pooled median of −0.0053, again negative in every group.

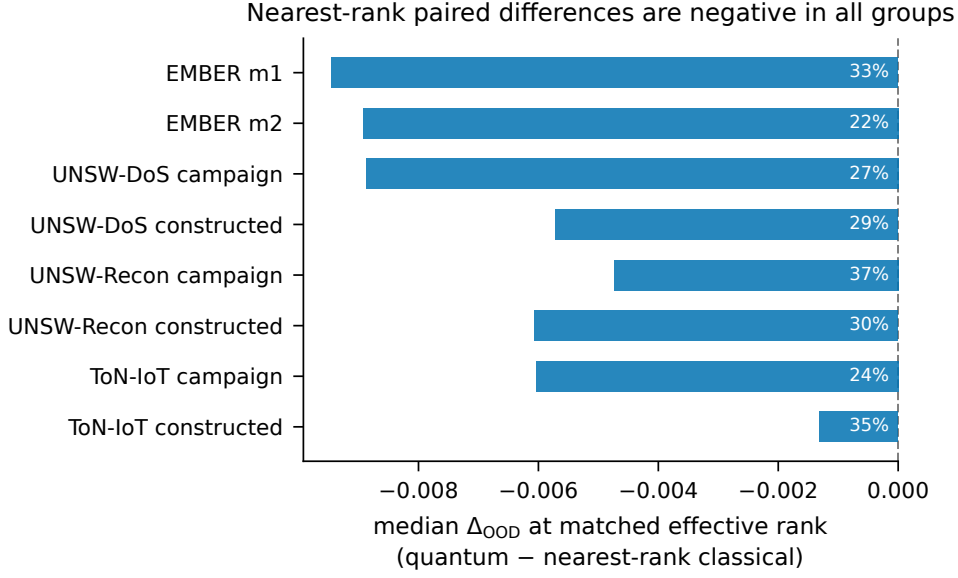


Fig. 4 Nearest-rank pairing yields negative median quantum-minus-classical differences in all eight fixed scenario-groups. Each quantum configuration is paired, with replacement, to the nearest-effective-rank classical configuration within run and dimension, subject to rank ratio ≤ 1.25 . The filter retains 48,708/64,800 pairs (75.2%; 69.3–83.3% by group). Bars show per-group median OOD balanced-accuracy differences; annotations give the fraction of retained pairs in which the quantum member is more accurate. This is observational matching; classical candidates may be reused and no causal family effect is implied.

Results remain negative over calipers 1.10, 1.50, 2.00, and without a caliper (Supplementary Table 3). These pairings describe the predefined region of rank overlap; they do not identify a causal effect of kernel family.

2.5 The quantum pool mixes entangling and product kernels

Circuit decomposition changes the interpretation of the aggregate family. The two ZZFeatureMap variants contain all-to-all two-qubit Pauli interactions, whereas the Pauli-X/Z and ZFeatureMap variants contain only one-qubit Pauli strings. For the latter pair, the prepared state and its fidelity kernel factorize over coordinates (Methods), permitting direct classical evaluation as a product of one-qubit overlaps. Thus two

Table 2 Logical circuit audit for the four quantum feature maps. Depth and CX columns give the values at 4 and 12 qubits after Qiskit 2.3.1 decomposition to **rz**, **sx**, **x**, and **cx**, optimization level 0, without device routing. “Fidelity CX” counts the compute–uncompute template for one kernel entry. Complete values at all five dimensions and one-qubit gate counts are provided in Supplementary Table 13 and the machine-readable artifact. These logical counts are not hardware-execution estimates.

Map	Pauli strings	Stratum	Product factorization	map depth (4/12)	map CX (4/12)	fidelity CX (4/12)
ZZ, 1 rep.	Z, ZZ	entangling ZZ	no	19/67	12/132	24/264
ZZ, 2 reps.	Z, ZZ	entangling ZZ	no	35/107	24/264	48/528
Pauli-X/Z, 1 rep.	X, Z	separable product	yes	11/11	0/0	0/0
Z, 2 reps.	Z	separable product	yes	8/8	0/0	0/0

of the four map families cannot supply evidence that entanglement or hard quantum kernel estimation is responsible for their behaviour.

Table 2 reports logical, unrouted resource counts. At 12 qubits, one ZZFeatureMap state preparation requires 132 CX gates for one repetition and 264 for two; the corresponding compute–uncompute fidelity templates require 264 and 528 CX gates. Both product-map templates require zero CX gates at every studied dimension. Under P1’ selection, product maps supply 594/1,080 (55.0%) of the selected quantum SVC configurations and 348/1,080 (32.2%) of the GP selections. The composition alone does not compare predictive quality; the matched stratum analysis below does so at 30 candidates per stratum.

At matched 30-candidate budgets, the source-dataset-equal SVC effect is -0.00928 for the entangling-ZZ stratum and -0.00547 for the product-map stratum; the constituent source-dataset means range from -0.01158 to -0.00580 and from -0.01167 to $+0.00021$, respectively. The entangling-ZZ point estimate is negative in every fixed scenario-group (-0.01960 to -0.00132); product-map estimates range from -0.02245 to $+0.00571$. The full 60-candidate pool remains at -0.00566 . Thus the controlled result is not produced by pooling product maps with entangling circuits, and restricting the quantum family to ZZ interactions does not reveal a hidden positive family effect. Supplementary Table 8 reports every pointwise conditional interval; these post-review strata are fixed-case sensitivities, not new population tests.

2.6 Within-v4 sensitivities separate the evaluation bundle

The within-v4 factorial removes the generation boundary from the historical specification map. In the q1000 SVC stratum, the fixed- C , same-test-oracle, customary-reference, native-budget endpoint is +0.0131, whereas the train-CV, ID-validation, extended-reference, equal-count endpoint is -0.0047 , a descriptive contraction of 0.0178. The 16 three-source-dataset-equal endpoints range from -0.0084 to +0.0131 (Figure 5; Supplementary Table 10). Thus the sign change is present within one result generation and does not require the legacy-to-v4 boundary.

Averaging paired changes over the other factorial axes gives -0.0084 for fixed $C = 1$ to train-CV, -0.0079 for customary to extended reference, -0.0037 for same-test oracle to ID validation, and -0.0002 for native to equal-count budget. These finite-design averages are not independent or causal contributions. In particular, the mean regularization-by-reference difference-in-differences is +0.0139, whereas the other five mean pairwise interactions have absolute magnitude at most 0.0034 (Supplementary Table 11). Regularization and reference strength must therefore be interpreted as part of an interacting evaluation bundle rather than as an additive attribution.

Removing the frozen port-, protocol-, and service-related fields also changes the relative comparison, confirming that common feature access alone does not make shortcut use family-invariant. Across the two network source datasets, the controlled effect moves from -0.0031 to -0.0245 (change -0.0215 ; source-dataset changes -0.0447 for ToN-IoT and +0.0018 for UNSW-NB15). The six group changes range from -0.0794 to +0.0099, and every pointwise change interval includes zero (Supplementary Table 12). The large negative change is concentrated in the ToN-IoT campaign shift, where removing direct traffic descriptors reduces both families' absolute accuracy but the selected fidelity kernel more strongly. This sensitivity neither restores a positive aggregate quantum effect nor establishes that all shortcuts have been removed.

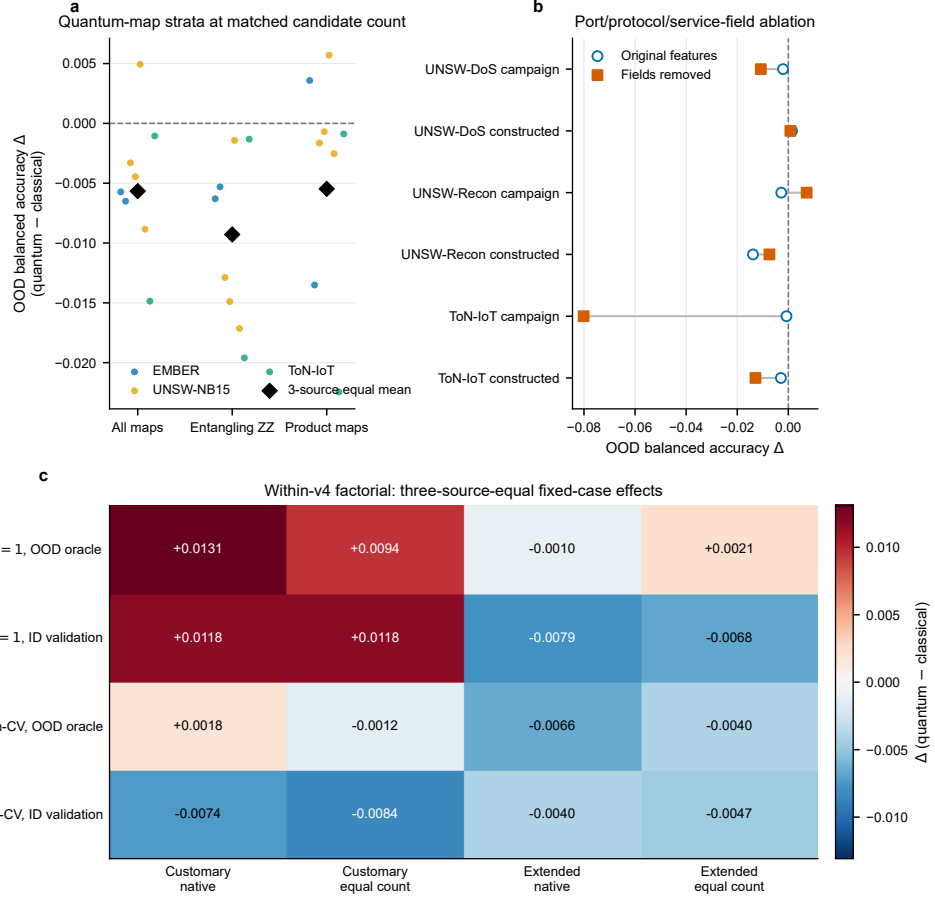


Fig. 5 Circuit-aware and within-v4 fixed-case sensitivities for SVC. **a**, Quantum-minus-classical OOD balanced-accuracy effects for all quantum maps ($B = 60$), entangling-ZZ maps ($B = 30$), and separable-product maps ($B = 30$), each against a blocked equal-count extended-classical reference. Coloured points are the eight scenario-groups and black diamonds are three-source-dataset-equal means. **b**, Original and ablated controlled effects after removing the frozen port-, protocol-, and service-related fields in the six q1000 network groups. Connecting lines show paired group changes. **c**, Three-source-dataset-equal effects for the q1000 within-v4 $2 \times 2 \times 2 \times 2$ factorial. Positive values favour the fidelity-kernel family. All quantities are descriptive summaries conditional on the fixed datasets and finite candidate pools; panels do not show population uncertainty or causal attribution.

2.7 Protocol sensitivity is prospectively corroborated on external domain shifts

We next applied the prospectively frozen protocol to College Scorecard, diabetes readmission, and ACS income, preserving the published TableShift domain splits and

withholding all OOD labels from the controlled selector. The complete external grid contains 30 task-size-seed units and 37,800 split-level results; every prespecified unit passed the automated author-provided audit. Figure 6 shows all task-size effects and conditional seed-cluster intervals.

The primary SVC result satisfies the machine-readable outcome category frozen as `replicated_protocol_sensitivity`; narratively, protocol sensitivity is prospectively corroborated across three fixed external tasks. The task-equal controlled effect is -0.0119 (95% conditional seed-cluster interval $[-0.0205, -0.0033]$, $n = 5$ deterministic subsampling seeds per task), whereas the weak-reference, fixed- C , same-test-oracle effect is $+0.0089$ ($[+0.0047, +0.0137]$). Their contraction is $+0.0208$ ($[+0.0113, +0.0307]$), with a leave-one-task-out range of $[+0.0116, +0.0260]$. Size-equal controlled point estimates are -0.0160 for College Scorecard, -0.0066 for diabetes readmission, and -0.0131 for ACS income; their pointwise intervals and both nested sizes are shown in Figure 6.

The secondary GP classifier shows a larger contraction of $+0.0324$ ($[+0.0223, +0.0427]$), but its task-equal controlled effect is $+0.0077$ with an interval spanning zero ($[-0.0015, +0.0183]$). College Scorecard leans quantum under this classifier, diabetes leans classical, and ACS is slightly positive. This classifier dependence limits the conclusion appropriately: the external study corroborates the sensitivity of the family verdict to evaluation design, not universal classical dominance.

2.8 Geometry is associated with robustness only partially

The results are consistent with both families reaching similar informative geometries. Geometry carries partial information about behaviour under shift, but neither the analysis nor the matching identifies it as a causal mediator: the association is regime-dependent, not the single governing law that optimistic accounts (our own earlier report included) proposed.

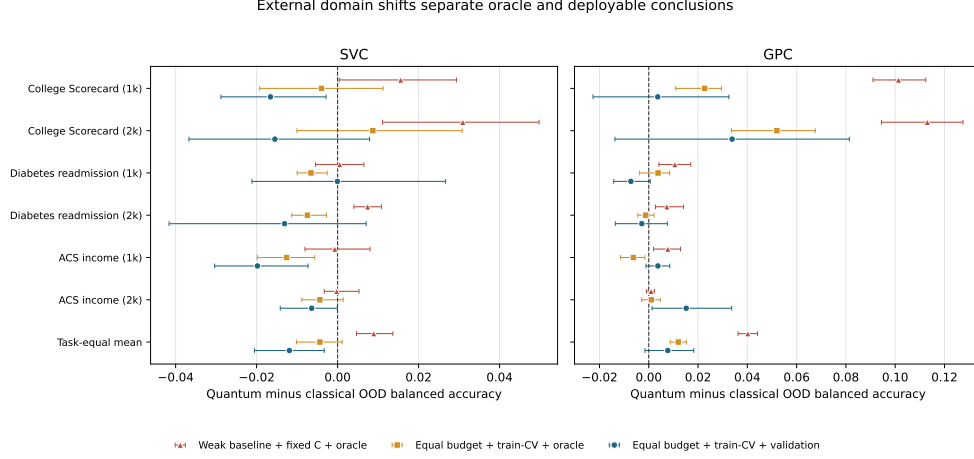


Fig. 6 External domain shifts separate optimistic and deployable conclusions. Effects are quantum minus classical OOD balanced accuracy for the weak-reference fixed-regularization same-test oracle, the equal-budget train-tuned same-test oracle, and the primary equal-budget train-tuned validation selector. Rows show both nested sizes for all three fixed tasks plus the task-equal mean. Error bars are 95% conditional seed-cluster intervals over five deterministic subsampling seeds per task ($n = 5$), preserving the nested-size relation. They quantify pipeline variation conditional on the fixed tasks and design, not uncertainty over a population of tasks.

The geometry- OOD link is a regime-dependent association, not a law.

Two descriptors of the training and OOD Gram matrices carry information about which kernels generalize: the effective rank of the training kernel and the centered kernel-target alignment on the OOD split. But their association with OOD accuracy is regime-dependent, not a single mechanism. Within scenario-groups (Figure 7 and Supplementary Table 4), the median Spearman correlation between effective rank and OOD accuracy ranges from $+0.80$ (EMBER $m2$) down to -0.15 (UNSW-NB15 reconnaissance under the constructed shift); partial correlations that control for family label and length scale track the raw values closely. OOD alignment is more consistently positive ($+0.22$ to $+0.64$) but likewise varies by regime. Because OOD KTA uses the labels of the same OOD split on which accuracy is measured, it is a post hoc, label-reusing diagnostic rather than an independently validated mechanism. A geometry model trained on two source datasets and used to rank kernels in the third achieves

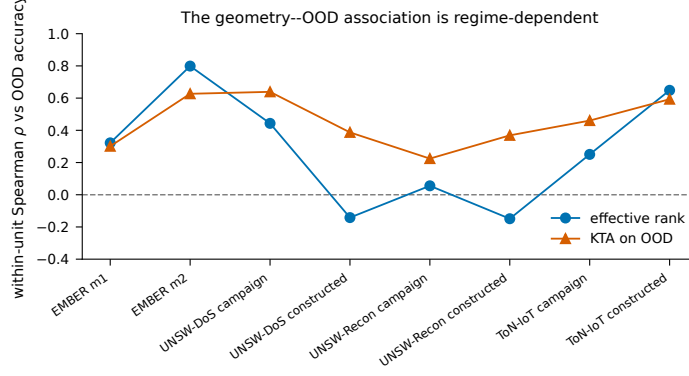


Fig. 7 The geometry--OOD association is regime-dependent. Points are median within-unit Spearman correlations between each geometry descriptor and OOD balanced accuracy, per fixed scenario-group for SVC; each unit is a run-dimension cell. Effective rank orders accuracy strongly in some regimes (EMBER $m2$, ToN-IoT $m2$) and weakly or negatively in others (the UNSW constructed-shift scenarios); OOD alignment is more consistently positive but never uniform. These descriptive associations have no population interval and do not identify a causal mechanism.

only Spearman 0.22–0.40 (EMBER 0.28, UNSW-NB15 0.22, ToN-IoT 0.40). We therefore describe a genuine but partial association, not a causal mechanism or deployable predictor.

2.9 Finite-shot estimation: the geometry is a statevector idealization

All quantum kernels above are computed by exact statevector simulation, which isolates kernel geometry but is not what a device would return. We therefore froze a post-confirmatory Monte Carlo sensitivity before computing its outcomes, then froze a reviewer-motivated out-of-sample correction before computing that extension. The subset contains one $q1000$ run per scenario-group, all at master, q-split, and model seed 42; within each run, the exact $P1'$ -selected quantum SVC configuration is fixed. For each of four shot counts, 30 independent measurement replicates replace every fidelity entry by a binomial estimate and reselect C from training data under three conditions: no PSD correction, independent projection of square train and OOD matrices

783 with rectangular blocks unchanged, and a coherent train-eigenspace Nyström exten-
784 sion (Methods). Stable SHA-256 seeds are distinct by run, configuration, shot count,
785 replicate, and matrix block.
786

788 One case-replicate samples 1,624,250 input-pair fidelities: 499,500 train-triangle
789 entries, 500,000 combined ID-holdout-train entries, 500,000 OOD-train entries, and
790 124,750 OOD-triangle entries. The ID-validation and ID-test roles partition the 500-
791 row holdout and therefore need not each contain exactly 250 rows. This corresponds
792 to 207,904,000 shots at 128 shots per fidelity and 13,305,856,000 at 8,192. Across eight
793 cases, there are 12,994,000 case-conditioned parameterizations; repeating them 30
794 times gives 389,820,000 fidelity-estimation instances per shot level. Estimating all eight
795 fixed configurations by circuits at all four levels projects 4.24 trillion shots (Supple-
796 mentary Table 14). Four configurations are separable-product maps, however, so direct
797 classical factorization can bypass their circuit sampling; the corresponding projection
798 for the four entangling cases alone is 2.12 trillion shots. The three PSD conditions
799 reuse the same sampled measurements. These are algebraic projections before device
800 routing, calibration, mitigation, and repeated job overhead, not executed hardware
801 costs or minimum implementation costs.
802

811 Table 3 summarizes the eight conditional Monte Carlo distributions. Sampling
812 does not induce a uniform accuracy direction. Under the original independent-square
813 heuristic, the median across the eight fixed scenario-group medians is -0.0005 at 128
814 shots (group range $[-0.0252, +0.0131]$), -0.0051 at 512, -0.0052 at 2,048, and 0.0000
815 at 8,192 shots. The coherent Nyström values are -0.0010 , -0.0026 , 0.0000 , and 0.0000 ;
816 the unprojected values are -0.0044 , -0.0009 , -0.0005 , and 0.0000 . PSD handling is
817 therefore part of the estimand rather than an innocuous cleanup step, but no condition
818 produces a uniform accuracy direction. Central 95% Monte Carlo intervals for all runs
819 are shown in Supplementary Figure 2.
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Table 3 Repeated finite-shot fidelity-estimation sensitivity for eight fixed exact Gram matrices (SVC). Each row summarizes, across scenario-groups, the within-run median over 30 measurement replicates. Δ_{OOD} is perturbed minus exact OOD balanced accuracy; brackets give the range of the eight medians, not a confidence interval. “Independent PSD” projects train and OOD square matrices separately while leaving evaluation–train blocks unchanged; it is an algorithmic heuristic, not one coherent out-of-sample kernel. “Nyström PSD” derives all corrected blocks from the sampled training matrix’s positive eigenspace. Rank ratio is perturbed/exact effective rank after clipping negative spectral mass and is shared across conditions. Per-run central 95% Monte Carlo intervals are in Supplementary Figure 2.

Shots	Condition	median Δ_{OOD} [group range]	median $ \Delta_{\text{OOD}} $	rank ratio
128	Unprojected	−0.0044 [−0.0563, +0.0137]	0.0118	1.93
128	Independent PSD	−0.0005 [−0.0252, +0.0131]	0.0070	1.93
128	Nyström PSD	−0.0010 [−0.0227, +0.0140]	0.0068	1.93
512	Unprojected	−0.0009 [−0.0226, +0.0192]	0.0068	1.48
512	Independent PSD	−0.0051 [−0.0281, +0.0041]	0.0084	1.48
512	Nyström PSD	−0.0026 [−0.0239, +0.0048]	0.0069	1.48
2,048	Unprojected	−0.0005 [−0.0222, +0.0055]	0.0080	1.24
2,048	Independent PSD	−0.0052 [−0.0238, +0.0041]	0.0072	1.24
2,048	Nyström PSD	0.0000 [−0.0225, +0.0093]	0.0065	1.24
8,192	Unprojected	0.0000 [−0.0158, +0.0114]	0.0101	1.13
8,192	Independent PSD	0.0000 [−0.0149, +0.0143]	0.0081	1.13
8,192	Nyström PSD	0.0000 [−0.0158, +0.0162]	0.0070	1.13

Effective rank is measurement-sensitive: the median perturbed-to-exact ratio falls from 1.93 at 128 shots to 1.13 at 8,192. Median absolute OOD-accuracy deviations are 0.0065–0.0069 under the Nyström extension and 0.0070–0.0084 under the independent-square heuristic; non-monotonicity is expected because each perturbed training matrix reselects a discrete C . OOD-KTA changes under Nyström are not directly comparable to those from the independently sampled OOD square because Nyström derives that square from OOD–train features; we report them as estimator-dependent diagnostics in the artifact. These distributions quantify measurement-sampling variability conditional on eight fixed exact Gram matrices. They do not model a device, circuit compilation, gate or readout noise, mitigation, or hardware feasibility.

2.10 Predictive uncertainty under shift

The Laplace GP probabilities provide a secondary uncertainty check (Supplementary Figure 1). For the target-label-free selected candidate, predictive entropy rises from

ID to OOD for both families in every scenario-group, while expected calibration error changes moderately and comparably. Together with the controlled accuracy results, these diagnostics provide no consistent evidence in the fixed cases of a calibration benefit unique to the selected fidelity kernels. We do not interpret rising entropy alone as evidence of calibration.

3 Discussion

The results support a narrow conclusion: the apparent family advantage is evaluation-sensitive. Under the controlled endpoint we observe no consistent evidence of a quantum advantage across the fixed low-qubit cases, while nearest-rank pairing yields negative median differences throughout the predefined overlap. Geometry descriptors carry partial, regime-dependent information and occur in both families. The quantum-specific question is sharpened by the circuit audit: half of the map blocks are separable product constructions, yet the matched entangling-ZZ stratum is more negative than either the product stratum or the full pool. The controlled direction therefore does not arise from pooling product kernels with entangling circuits.

From the perspective of distribution-shift research, this reinforces that robustness conclusions are conditional on the protocol [11–13, 28]. Within v4, moving from the fixed- C , same-test-oracle, customary-reference, native-budget corner to the fully controlled corner changes the q1000 estimate from +0.0131 to -0.0047 . Train-CV regularization and the extended reference have the largest mean paired changes, but their sizeable interaction precludes an additive or causal reading. The legacy/v4 display still cannot isolate regularization, and the larger historical +0.037 endpoint remains interpretable only as a complete evaluation bundle.

The external analysis provides prospective corroboration beyond the constructed security shifts. Under a protocol frozen before result inspection, three education, health, and socioeconomic tasks corroborate a positive protocol contraction and a

negative controlled SVC effect. The result is stable to omitting any one task, yet heterogeneous by task and classifier: the GP endpoint retains a small positive task-equal lean whose interval crosses zero. The strongest defensible conclusion is therefore about evaluation sensitivity, not the universal ordering of kernel families.

Length scale is decisive for both classical and quantum kernel behaviour [29, 30]. With symmetric scale freedom and tuned regularization, the families occupy overlapping effective-rank and alignment regimes. These quantities are kernel descriptors, not unique signatures of quantum provenance; OOD alignment additionally reuses target labels and remains strictly post hoc.

The high-rank finding must also be reconciled with two spectral results that superficially point the other way. Kübler et al. argue that quantum kernels generalize in-distribution only when the relevant spectral mass is concentrated in few components [31], and Thanasilp et al. show that fidelity kernels concentrate exponentially—toward effectively rank-one, uninformative Gram matrices—as qubit counts grow [32]. There is no contradiction, but there is a genuine tension that our results locate precisely. Both results concern the asymptotic or i.i.d. regime; our effective ranks (roughly 10–150 on training sets of 10^3 – 4×10^3 samples at 4–12 qubits) sit far below the concentration regime. The repeated finite-shot sensitivity adds an empirical caveat these theoretical results anticipate: effective rank from sampled fidelities is inflated by a median factor of 1.93 at 128 shots and 1.13 at 8,192 shots (Section 2.9). Exact effective rank is therefore a statevector idealization even at these modest qubit counts.

From a practical perspective, no consistent fidelity-kernel advantage is observed over the tuned classical family, while overlap estimation introduces circuit and shot costs absent from direct classical-kernel evaluation. A shortened-length-scale Laplacian is therefore an important inexpensive reference, not a universally superior choice. More generally, comparisons should keep representation and learner fixed, tune all regularization on training data, match search budgets, select without target labels, include a

967 non-constructed shift where possible, and report correlated benchmarks as fixed cases.
968
969 Geometry descriptors remain useful diagnostics, but not deployable selectors unless
970 validated without target-label access.
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972 Several limitations define the scope of our conclusions.

973 Our shift operationalization covers three reproducible mechanisms—within-class
974 sparsity-tail extremes, train-geometry-tail extremes, and a held-out attack-campaign
975 shift on network traffic (an unseen attack campaign plus a capture-partition change,
976 not a timestamped temporal split)—but not the full space of real-world drift processes.
977
978 Explicit temporal splits on timestamped malware, family-based drift, and adversarially
979 induced shift remain outside the study’s scope, and our conclusions should be re-tested
980 under them rather than assumed.
981

982 Kernel and encoding coverage also remains finite. The classical family spans seven
983 kernels plus a bandwidth sweep, and both families receive symmetric tuning freedom,
984 but neither candidate set is exhaustive: the quantum side is restricted to four com-
985 monly used fidelity feature maps at low qubit counts, and trained or task-adapted
986 encodings, projected kernels, and deeper maps are not evaluated. The observed asso-
987 ciation suggests that future extensions should report their Gram-matrix geometry as
988 well as provenance, but it does not predict their outcome. At substantially larger
989 qubit counts, exponential kernel concentration [32] creates an additional regime not
990 represented here, so our conclusions should not be extrapolated beyond the low-qubit
991 study.
992

993 All selection procedures operate over finite candidate sets. Candidate matching,
994 full-pool endpoints, and budget curves expose part of this dependence, but conclu-
995 sions can still change with candidate definitions and search granularity. Fully nested
996 selection over additional independent benchmark pools would strengthen the analysis.
997

998 The quantum kernels are simulated rather than executed on hardware. The study
999 therefore characterizes fidelity-kernel geometry, not noisy-device behaviour. Repeated
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binomial sampling quantifies one component of measurement uncertainty, but hardware noise, transpilation, readout, mitigation, and circuit-dependent correlations remain unmodelled [33]; exponential concentration may also make informative spectral structure expensive to resolve at larger scales [32]. We make no claim about hardware feasibility, resource advantage, or quantum speedup—the only rigorously established quantum-classical learning separation concerns a specifically constructed problem, not natural data [34]. At the dimensions studied, the fidelity kernels are also classically simulable [35, 36], which is precisely what makes them available as practical kernel constructions today.

Dataset scope and feature caveats provide a final boundary. Dynamic malware analysis and encrypted traffic remain untested. The network exports retain port-derived counters and, for ToN-IoT, direct port, protocol, and service indicators that can support shortcuts in NIDS benchmarks [37]. Sharing features controls the inputs but does not imply that distinct kernels exploit them equally. In the frozen q1000 ablation, removing these fields shifts the two-source-dataset-equal effect from -0.0031 to -0.0245 , but the change is highly heterogeneous and concentrated in ToN-IoT. This is evidence that feature access can affect the relative comparison, not proof that every shortcut has been removed or that the ablated tasks define a preferred deployment benchmark.

We do not claim universal superiority, equivalence, complexity-theoretic advantage, hardware relevance, or a refutation of quantum kernels. The open problem for both families is whether a label-free, train-only signal can identify a useful geometry under shift; label-free drift detection [38] and task-model alignment theory [39–41] provide starting points. This interpretation is consistent with calls for controlled QML benchmarking [6–8].

1059 4 Methods

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1062 4.1 Problem setting and estimands

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1064 We study binary security classification—malware versus benign software and attack
1065 versus benign traffic—under distribution shift. Let D_{ID} denote the in-distribution
1066 process and D_{OOD} a constructed stress-test or held-out-campaign process. Each run
1067 contains a training set T , an ID holdout that is divided into validation V_{ID} and test
1068 S_{ID} , and an OOD test set S_{OOD} . Preprocessing, samples, classifier implementation,
1069 tuning rule, and family-selection rule are shared; only the kernel changes. Comparisons
1070
1071 are always within a classifier.

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1073 The primary endpoint is OOD balanced accuracy after P1' deployment selection.

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1075 A family effect is

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$$\Delta_{\text{OOD}} = \text{BAcc}_{\text{OOD}}^{(Q)} - \text{BAcc}_{\text{OOD}}^{(C)},$$

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1079 so positive values favour the quantum family. ID balanced accuracy and ID-to-OOD
1080 degradation are secondary. P2 selects from OOD information in other realizations,
1081 and P3 selects and evaluates on the same OOD test labels; both are explicitly non-
1082 deployable descriptive oracles.

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1085 4.2 Security datasets and shared split sizes

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1087 The source datasets are EMBER [15], UNSW-NB15 [18], and ToN-IoT [19]. Four
1088 binary scenarios—EMBER malware, UNSW-NB15 DoS, UNSW-NB15 Reconnaissance,
1089 and ToN-IoT Scanning—produce two shift mechanisms each and hence eight
1090 scenario-groups. Classical and quantum kernels always receive identical row indices
1091 and class balance.

1092

1093 The 72 principal settings cross eight groups with master seeds $\{42, 123, 999\}$ and
1094 $(n_{\text{train}}, n_{\text{ID}}, n_{\text{OOD}})$ equal to $(1000, 500, 500)$, $(2000, 1000, 1000)$, or $(4000, 1800, 1800)$.

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1096 Each setting is instantiated with q-split seeds $\{42, 123, 999, 7, 2024\}$ and model seeds

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{42, 123, 999}, yielding fifteen pipeline realizations arranged in five q-split clusters. Indices are stored explicitly; automated gates verify split-role disjointness, required cardinalities, both classes, identical family inputs, and absence of label-derived features.

4.3 EMBER shift construction

EMBER uses the histogram and byte-entropy feature blocks and two label-conditional tail stress tests. They are not described as certified covariate shifts because conditioning the split on class does not establish invariance of $P(Y | X)$. For $m1$, the sparsity score is the number of non-zero entries in the selected feature block. Within each class, low- and high-score extremes supply the exact-size OOD set; training and ID samples are stratified draws from the remainder.

For $m2$, the final training set is fixed first. A MaxAbs scaler and TruncatedSVD representation are fitted on training data only, and a centroid is computed for each class. Every non-training sample is scored by Euclidean distance to its own-class centroid. Within-class low- and high-distance extremes form S_{OOD} , and the ID holdout is sampled from the remaining pool. This construction probes unusual geometry relative to the fixed training set without using test information to fit the representation.

4.4 Network-flow shift construction

Numeric preparation removes identifiers and explicit label derivatives, leaving 39 UNSW-NB15 and 88 ToN-IoT features. The UNSW-NB15 export excludes direct categorical protocol, service, and state fields but retains three port-derived counters; ToN-IoT retains two port fields plus protocol and service indicators. A leakage audit records every exclusion and feature-label association; the largest observed absolute correlation is 0.62 and belongs to a retained traffic feature. Each scenario has reference and current pools and uses two mechanisms.

1151 The *m2-centroid* mechanism is the network analogue of EMBER *m2*: train-only
1152 MaxAbs+TruncatedSVD class centroids define within-class distance tails, which sup-
1153 ply the OOD set. The *attack-campaign* mechanism performs no geometric selection.
1154 Training and ID pools contain benign traffic and all non-target attack categories;
1155 OOD contains benign traffic and the held-out target campaign. UNSW-NB15 refer-
1156 ence and OOD pools additionally use different official capture partitions. This is an
1157 unseen-campaign plus capture-partition shift, not a timestamped temporal split.

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1164 4.5 Shared preprocessing and representations

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1166 For every run and dimension $d \in \{4, 6, 8, 10, 12\}$, a MaxAbs scaler and TruncatedSVD
1167 are fitted on T ; the reduced coordinates are standardized and mapped linearly with
1168 training-fitted parameters so that the *training* coordinates lie in $[0, \pi]$. Held-out coor-
1169 dinates receive the same affine map and may extrapolate outside that interval under
1170 shift. The same embedded samples are supplied to both families, making the compar-
1171 ison a kernel swap rather than a representation swap. Dimension is a candidate axis.
1172 No validation, ID-test, or OOD-test row fits a transformation. The SVD random state
1173 equals the model seed.

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1181 4.6 Classifiers and train-only regularization

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1183 The primary learner is an SVC with a precomputed Gram matrix and balanced class
1184 weights [42]. For every kernel geometry, $C \in \{0.01, 0.1, 1, 10, 100\}$ is selected by five-
1185 fold stratified cross-validation on the training Gram matrix; ties favour the smaller C .
1186 When $n_{\text{train}} > 2000$, this selection step uses a class-stratified random-state-20260717
1187 subsample capped at 2,000 rows, while the final model is always refitted on the com-
1188 plete training block. Because C is resolved before family selection, its grid points are
1189 not counted as separate family candidates.

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The secondary learner is a binary Gaussian-process classifier with a logistic link and Laplace approximation [43]. It consumes the same precomputed kernel, uses diagonal jitter 10^{-8} , at most 100 Newton iterations, and convergence tolerance 10^{-6} . The logistic-probit approximation supplies predictive probabilities; no post-hoc calibration is applied.

4.7 Classical kernel family

The customary reference contains a cosine-normalized linear kernel and RBF kernel. The extended family adds cosine-normalized degree-2 and degree-3 polynomial kernels with $\gamma = 1/d$ and intercept 1, the Laplacian kernel $\exp(-\|x-x'\|_1/\ell_1)$, and Matérn-3/2 and Matérn-5/2 kernels. The base ℓ_1 and ℓ_2 scales are median pairwise Manhattan and Euclidean distances, respectively, calculated from at most 2,000 training rows with the model seed; the RBF base scale is fitted from training variance. RBF, Laplacian, and both Matérn kernels use factors $\{0.1, 0.3, 1, 3, 10\}$ around these training-only bases. Deduplication yields 23 kernel-shape/scale blocks crossed with five dimensions, or 115 extended-classical candidates in full-coverage groups.

4.8 Quantum fidelity-kernel family

The quantum family uses exact statevector fidelities

$$K_Q(x, x') = |\langle \phi_Q(x) | \phi_Q(x') \rangle|^2$$

[1, 2]. Qiskit 2.3.1 implements four map blocks [44]: fully connected ZZ maps with one and two repetitions; a one-repetition Pauli map with single-qubit strings X and Z; and a two-repetition Z map. Only the ZZ maps contain two-qubit interactions. The latter

1243 two prepare product states,

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1250 so the constructor argument `entanglement="full"` stored for the Pauli-X/Z map has

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1252 no effect without a multiqubit Pauli string. The product can be evaluated classically

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1254 in $O(d)$ one-qubit-overlap calculations. We call these the entangling-ZZ and separable-

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1256 product strata. Each map uses angle scales $\{0.5, 1, 2\}$ and all five dimensions, giving

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1258 60 candidates (30 per stratum) at 4–12 simulated qubits. Classical bandwidth and

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1260 quantum angle-scale freedom both control locality [29, 30]; no circuit is trained.

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1261 4.9 Circuit and sampling-resource audit

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1265 Circuit counts use Qiskit 2.3.1 and the logical basis $\{\mathbf{rz}, \mathbf{sx}, \mathbf{x}, \mathbf{cx}\}$ at transpiler

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1267 optimization level 0, without backend layout or routing. We report each feature-map

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1269 template and the compute–uncompute fidelity template formed by composing the map

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1271 at x with the inverse map at x' after assigning distinct numeric inputs. Counts are

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1273 compiler- and basis-conditional logical resources, not device schedules.

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1275 For the repeated q1000 finite-shot sensitivity, each case and measurement replicate

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1277 estimates the off-diagonal training triangle, the combined 500-row ID-holdout-to-

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1279 training rectangle (partitioned downstream into validation and test roles), the

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1281 OOD-test-to-training rectangle, and the off-diagonal OOD square triangle; diagonals

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1283 are fixed to one. Thus one kernel configuration requires

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$$\binom{1000}{2} + 500(1000) + 500(1000) + \binom{500}{2} = 1,624,250$$

distinct fidelity estimates per shot level and replicate. Multiplying this count by shots
gives the idealized circuit-shot total. The three PSD conditions reuse the same sam-
pled entries and add only classical post-processing. Routing, calibration, mitigation,
queueing, and device noise are excluded.

4.10 Target-label-free family selection

The original ID holdout is divided deterministically with class stratification into
 V_{ID} and S_{ID} . Within each class, global row indices are ordered by SHA-256 of the
salt `ksf-v4-idsplit::` and index, then assigned alternately to validation and test.
The halves are disjoint and approximately class-balanced. P1' deploys, within each
family, the candidate with highest V_{ID} balanced accuracy and reports it once on
 S_{ID} and S_{OOD} . Ties follow lexicographic candidate order. OOD labels never enter
regularization, preprocessing, or P1' selection.

4.11 Candidate-budget matching

Best-of-family selection rewards a larger pool. The confirmatory coverage audit found
115 extended-classical and 60 quantum candidates in every one of the eight scenario-
groups and both classifiers. The primary comparison therefore draws 60 candidates
from each family in all 16 group-classifier cells. For each cell, 5,000 extended-classical
subsamples are generated with seed 20260715. The primary `kernel_blocked` scheme
samples 12 of the 23 complete kernel-shape/scale blocks, each crossed with all five
dimensions, to mirror the 12 quantum feature-map/angle-scale blocks crossed with the
same dimensions. This matches both candidate count and block structure, although it
does not make the two search spaces identical. `uniform` samples individual candidates,
while `kernel_stratified` distributes the draw across kernel-shape/scale strata. Each
subsample selects its P1' winner; the classical endpoint is the resampling expectation.
The full 115-candidate pool is secondary. Resampling percentiles describe sensitivity
to the sampled candidate set, not population uncertainty.

1335 4.12 Entangling and separable quantum strata

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1337 The reviewer-motivated v0.8 sensitivity was committed before its performance con-
1338 trasts were calculated. It reuses all v4 summaries and compares three SVC pools under
1339 train-CV and P1': the full 60-candidate quantum family, the 30 entangling-ZZ candi-
1340 dates, and the 30 separable-product candidates. The 60-candidate pool is compared
1341 with 12 complete classical blocks and each 30-candidate stratum with six complete
1342 classical blocks, always crossed with all five dimensions. Each classical endpoint is the
1343 expectation over 5,000 `kernel.blocked` draws using SHA-256-derived seeds rooted at
1344 20260729. This tests dependence on map stratum, not computational advantage.

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1352 4.13 Within-v4 factorial and shortcut sensitivity

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1354 The within-generation SVC factorial uses all 360 q1000 runs and crosses: same-test
1355 OOD versus disjoint ID-validation selection; fixed $C = 1$ versus train-CV C ; cus-
1356 tomary versus extended classical reference; and native versus equal-count budgets.
1357 Fixed- C predictions are recomputed with the deterministic v4 preprocessing, kernel
1358 constructors, samples, and split roles; an integrity gate requires exact candidate-split
1359 key identity with every archived v4 SVC summary and the original ID-split hash salt.
1360 With the customary 30-candidate reference, equal count samples six of twelve quan-
1361 tum blocks; with the extended reference, it samples 12 of 23 classical blocks. The
1362 same 5,000 frozen block draws are paired across selection and regularization cells. Six-
1363 teen fixed-case endpoints and their paired axis contrasts are descriptive; they do not
1364 identify causal effects.

1372 The q1000 shortcut sensitivity uses the same design on all 270 network runs
1373 after removing features before the train-only embedding. UNSW-NB15 excludes
1374 `ct_src_dport_ltm`, `ct_dst_sport_ltm`, and `is_sm_ips_ports`; ToN-IoT excludes
1375 source/destination ports, all `proto_*`, `service`, and `service_*` fields. This removes 3
1376 of 39 and 14 of 88 fields, respectively. The manuscript-facing contrast uses train-CV,
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P1', and 60-candidate block matching with paired draws. It is a port/protocol/service-field ablation, not a guarantee that every shortcut has been removed.

4.14 Security fixed-case aggregation and conditional intervals

The effective replication axis within a scenario-group is q-split seed. Master seed is a design stratum rather than a replicate, some EMBER OOD pools recur across master seeds, and training sizes are nested. Audited OOD overlap is at most 1.2% for EMBER and 7.8% for network groups. Within each q-split cluster, model seeds are averaged within master seed, master seeds within size, and the three sizes with equal weight. The five resulting cluster means T_j give the group effect $\bar{T}$ and 95% conditional cluster- t interval

$$\bar{T} \pm t_{4,0.975} \frac{s_T}{\sqrt{5}}.$$

This few-cluster t construction follows the cluster-mean approach of Ibragimov and Müller [45]. The interval ($n = 5$ q-split clusters) is conditional on the fixed benchmark pools and experimental design; inclusion of zero is not evidence of equivalence. A 9,999-draw BCa bootstrap over cluster means and leave-one-q-split calculations are secondary diagnostics, reported for all 16 primary cells in Supplementary Table 6. The five cluster values behind every interval are exposed in Supplementary Table 7.

Across groups, EMBER $m1/m2$ are averaged within source dataset, the four UNSW-NB15 groups within UNSW-NB15, and the two ToN-IoT groups within ToN-IoT. These three source-dataset means receive equal weight. Leave-one-source-dataset-out ranges omit one of the three at a time. The eight scenario-groups and three datasets were not sampled from defined populations; therefore no population p -value or dataset-population confidence interval is reported.

1427 4.15 Specification curve

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1429 The S1–S10 display is retained as a historical cross-generation sensitivity map over
1430 complete legacy and v4 endpoints. S1–S4 use fixed- C legacy generation and S5–S10
1431 train-CV v4 generation, so their boundary cannot isolate regularization. The rows
1432 remain in their prespecified order and are never sorted by effect. Individual-axis inter-
1433 pretation instead uses the within-v4 factorial above. All summaries average realizations
1434 within setting and group, then groups within the three source datasets, and finally
1435 source datasets with equal weight; no specification-level hypothesis test is performed.
1440

1442 4.16 Prospectively frozen external corroboration

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1444 Before acquiring TableShift data or inspecting any external result, we committed
1445 the tasks, sampling, preprocessing, candidate pools, estimands, uncertainty proce-
1446 dure, and outcome-category rule. The source implementation is pinned to commit
1447 `fca9429814703a07e3902d005d46563a207b7f0a` [46]. Three public tasks span distinct
1448 domains and published shift variables: College Scorecard (institution type), diabetes
1449 readmission (admission source), and ACS income (geographic region). TableShift’s
1450 `train`, `validation`, `id_test`, and `ood_test` roles are preserved; `ood_validation` is
1451 never loaded. College Scorecard is read from its public CC0 archive because the
1452 pinned downloader invokes authenticated Kaggle access; the pinned parser, schema,
1453 and domain split are unchanged. Source-file SHA-256 hashes and the container digest
1454 are recorded.

1462 Each task has nested strata $(n_{\text{train}}, n_{\text{val}}, n_{\text{ID}}, n_{\text{OOD}}) = (1000, 250, 250, 500)$ and
1463 $(2000, 500, 500, 1000)$. For seeds $\{42, 123, 999, 7, 2024\}$, source rows are ordered by
1464 SHA-256 of task, split, seed, and original position, giving deterministic label-blind
1465 samples. Gates verify source-role disjointness, size nesting, both classes, source-schema
1466 preservation, and at least 12 non-constant encoded features. All 30 task–size–seed
1467 units passed.

All transformations fit `train` only. Constant columns are removed; numeric variables receive median imputation; Boolean, categorical, and string variables receive most-frequent imputation and one-hot encoding with unknown categories ignored. MaxAbs scaling, TruncatedSVD at the five study dimensions, standardization, and a training-fitted affine map that places training coordinates in $[0, \pi]$ follow; held-out coordinates may extrapolate beyond that range. The SVD random state is 20260729. The security kernel and classifier pools are then reused.

For each task, size, seed, and classifier, the validation-selected 60-candidate quantum endpoint is compared with 5,000 `kernel_blocked` 60-candidate subsamples of the 115-candidate classical pool using seed 20260729. The primary effect is quantum minus expected classical OOD balanced accuracy. The equal-budget OOD oracle selects on `ood_test`; the weak-fixed oracle uses linear/RBF, fixes SVC at $C = 1$, and also selects on the reported OOD labels. Protocol contraction is weak-fixed-oracle effect minus controlled $P1'$ effect.

Seeds are averaged within task and size, nested sizes receive equal weight within task, and the three fixed tasks receive equal weight. Conditional 95% intervals use 9,999 resamples of the five seed clusters within task while preserving nested sizes. Leave-one-task-out ranges are descriptive. The intervals quantify deterministic-subsampling variability conditional on three fixed tasks; they do not represent a population of tasks.

4.17 Performance and uncertainty metrics

Balanced accuracy on S_{OOD} is primary. Accuracy, macro and positive-class F1, ROC AUC, and precision-recall AUC are recorded on every split. The GP analysis additionally records log loss, Brier score, expected calibration error with 15 equal-width probability bins [47], and mean predictive entropy. No threshold or calibration parameter is fitted on OOD labels.

1519 4.18 Geometry descriptors and association analyses

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1521 For positive-semidefinite K with eigenvalues λ_i and $p_i = \lambda_i / \sum_j \lambda_j$, effective rank is

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1528 [24]. For $y \in \{-1, +1\}^n$ and doubly centered K_c , centered kernel-target alignment is

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1535 [25, 26]. KTA is computed separately on training, ID, and OOD square blocks, along-

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$$r_{\text{eff}} = \exp\left(-\sum_i p_i \log p_i\right)$$

$$\text{KTA}(K, y) = \frac{\langle K_c, yy^\top \rangle_F}{n \|K_c\|_F}$$

Within each run–dimension cell, Spearman correlations relate effective rank, OOD KTA, and KTA survival (OOD minus ID) to SVC OOD balanced accuracy for all, classical-only, and quantum-only candidates. Partial Spearman association residualizes ranked variables against family and log length/angle scale. For portability, a descriptive model bins log effective rank into ten training-dataset quantiles, learns mean OOD accuracy per bin from two source datasets, and ranks cells in the third; held-out Spearman correlation is reported once per source dataset.

Rank matching is observational. Within each run and dimension, every quantum candidate is first paired to the classical candidate with smallest absolute effective-rank difference; classical candidates may be reused. Rank ratio $\max(r_Q, r_C) / \min(r_Q, r_C)$ is filtered at prespecified calipers 1.10, 1.25, 1.50, and 2.00, plus unfiltered reporting. A sensitivity uses minimum-cost one-to-one assignment with absolute log-rank difference, leaving excess candidates unmatched. Counts, retained fractions, rank discrepancies, mean and median OOD differences, and the fraction favouring quantum are reported for every caliper. None of these analyses identifies a causal geometry or family effect.

4.19 Repeated finite-shot perturbation

The original v0.6.0 finite-shot analysis and the reviewer-motivated v0.7.0 out-of-sample PSD extension were each frozen before their outcomes were computed. They fix one $q1000$, master-seed-42, q-split-42, model-seed-42 run per security group and, within each, the exact-statevector quantum SVC candidate selected by $P1'$ on V_{ID} . Lexicographic candidate order resolves ties. Input `summary_v4.csv` SHA-256 hashes, chosen kernels, dimensions, and exact C values are stored in the frozen specifications.

For shots $s \in \{128, 512, 2048, 8192\}$ and measurement replicates $r = 0, \dots, 29$, each exact fidelity p is sampled independently as $\text{Binomial}(s, p)/s$. A separate NumPy generator is used for train, ID-validation, ID-test, OOD-test, and OOD-square blocks. Its unsigned 64-bit seed is the first eight SHA-256 bytes of

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ksf-v6-shots::::<kernel>::::<shots>::::<block>.
```

Python’s process-randomized hash is not used.

Square train and OOD blocks are sampled on the upper triangle, mirrored, clipped to $[0, 1]$, and assigned unit diagonal; rectangular blocks receive binomial sampling and clipping. Three prespecified conditions use the same sampled blocks. The unprojected condition makes no PSD correction. The independent-square heuristic clips negative eigenvalues of the train and OOD square matrices separately while leaving every rectangular evaluation–train block unchanged; it is an algorithmic intervention for a precomputed SVC, not a coherent correction of one out-of-sample kernel.

The coherent Nyström condition eigendecomposes the sampled training matrix as $K_T = U \text{diag}(\lambda) U^\top$ and retains $\lambda_i > \tau$, where $\tau = \max\{10^{-12}, \epsilon n \max(1, \max_i |\lambda_i|)\}$. For retained U_+ and λ_+ ,

$$K_T^+ = U_+ \text{diag}(\lambda_+) U_+^\top, \quad \Phi_E = K_{E,T} U_+ \text{diag}(\lambda_+)^{-1/2}.$$

1611 The corrected evaluation–train and evaluation-square blocks are $\Phi_E \Phi_T^\top$ and $\Phi_E \Phi_E^\top$,
 1612 respectively, with $\Phi_T = U_+ \text{diag}(\lambda_+)^{1/2}$. Thus the OOD square used for KTA is
 1613 derived from the sampled OOD–train block rather than independently projected. In
 1614 all three conditions, C is reselected by the same five-fold train-only procedure and
 1615 the final SVC is refitted. Outputs include signed and absolute OOD change from the
 1616 fixed exact endpoint, effective-rank ratio, OOD-KTA change, retained rank, eigen-
 1617 value tolerance, negative-eigenvalue fraction, and Frobenius changes for every block.
 1618 Within each fixed Gram matrix, medians and central 2.5–97.5% Monte Carlo inter-
 1619 vals summarize 30 replicates. Across the eight fixed matrices, medians and ranges are
 1620 descriptive. This model contains no device or correlated-noise process.
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1629 4.20 Statistics and reproducibility

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 1631 No null-hypothesis test is used for the eight security groups, three source datasets,
 1632 or three external tasks. Every uncertainty object names its effective n , resampling
 1633 unit, and conditioning set: security intervals use five q-split clusters; external intervals
 1634 use five deterministic seed clusters per task; finite-shot intervals use 30 measurement
 1635 replicates conditional on one fixed Gram matrix. The fifteen security pipeline realiza-
 1636 tions are never treated as independent. Security intervals use the t_4 reference and are
 1637 pointwise, not simultaneous; the five underlying cluster means are published. Inclusion
 1638 of zero is never interpreted as equivalence. No multiplicity adjustment was prespec-
 1639 ified. Because multiple intervals and diagnostics are displayed, they are interpreted
 1640 descriptively rather than as a familywise error-controlled decision procedure; the BCa
 1641 analysis with five clusters is retained only as a fragile sensitivity.
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1649 Sample sizes were design- and compute-constrained and fixed before outcome anal-
 1650 ysis; no power calculation was used. No completed group, task, specification, budget
 1651 scheme, caliper, or finite-shot replicate was removed based on its result. Randomized
 1652 operations use recorded seeds; external sampling and ID division use deterministic
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SHA-256 ordering. The test suite validates source-dataset mapping, endpoint equality across analyses, candidate-budget coverage, matching uniqueness, seed stability, exact-reference hashes, and complete Monte Carlo cells.

4.21 Software environment

Experiments use Python 3.11/3.12, NumPy 2.3.5, pandas 2.2.3, scikit-learn 1.8.0, Matplotlib 3.11.0, and Qiskit 2.3.1. The Windows-specific Conda snapshot and cross-platform package specifications serve different archival purposes; neither is an exact machine image of every full-scale run.

4.22 Use of generative AI

OpenAI Codex was used during manuscript revision for language editing, code inspection, analysis implementation, and consistency checking. It did not determine authorship, study accountability, or acceptance of any scientific claim. The authors inspected the generated changes, verified all reported analyses and citations, and take full responsibility for the manuscript.

Declarations

Data Availability. This study uses the public EMBER [15], UNSW-NB15 [18], ToN-IoT [19], and TableShift [46] benchmarks, which remain subject to their original access conditions. Raw source records are not redistributed. Derived split definitions, deterministic row-position hashes, shift constructions, experiment manifests, complete aggregated results, and summary tables are available through the project Zenodo concept record at <https://doi.org/10.5281/zenodo.19147649>. The exact v0.7.0 submission artifact is identified by the immutable version DOI <https://doi.org/10.5281/zenodo.21676563> [48].

Code Availability. Code for constructing the splits, running the kernel experiments, reproducing every analysis, and regenerating every table and figure is available at https://github.com/roberto-fernandez-barrios/kernel_shift_framework and in the versioned Zenodo archive above. A Windows-specific Conda snapshot and platform-neutral package specifications document the software dependencies; neither is claimed as an exact machine image of every full-scale run. The code is released under BSD-3-Clause with no restriction on non-academic use.

Competing interests. All authors declare no financial or non-financial competing interests.

Ethics approval and consent to participate. This computational study uses only public benchmark releases and involved no participant recruitment, intervention, or access by the authors to direct identifiers. No new human data were collected, and no new participant consent or ethics approval was required for this secondary benchmark analysis. The original data providers' access conditions and governance remain applicable.

Authors' contributions. R.F.-B.: conceptualization, methodology, software, data curation, formal analysis, visualization, investigation, writing—original draft, and writing—review and editing. I.P.-L.: supervision, validation, and writing—review and editing. A.G.-S.: validation, project administration, and writing—review and editing. P.G.B.: supervision, resources, and writing—review and editing. All authors read and approved the final manuscript.

Acknowledgements. This study received no funding.

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